

Uncovering Patterns in Squirrel Behaviour

A Data Science Investigation into
Central Park's Wildlife

- **PRESENTED BY**

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Today's Agenda

Key topics and discussion points to be reviewed in today's presentation

- | | | | |
|----------------------|----|-------------------------|----|
| • Report Overview | 01 | • Insights Summary | 05 |
| • Research Questions | 02 | • Findings Report | 06 |
| • Research Method | 03 | • Research Limitations | 07 |
| • Result Analysis | 04 | • Conclusion & Insights | 08 |



Report Overview

01

TOPIC: EASTERN GREY SQUIRREL ADAPTATION IN URBAN ENVIRONMENTS

CONTEXT & DATASET

Dataset: 2018 Central Park Squirrel Census.

- 3,023 sighting records across 350 hectares.
- Captures behavioural, spatial, and environmental data.

Challenges: Severe class imbalance (only **5.9%** approached humans), purely observational data.

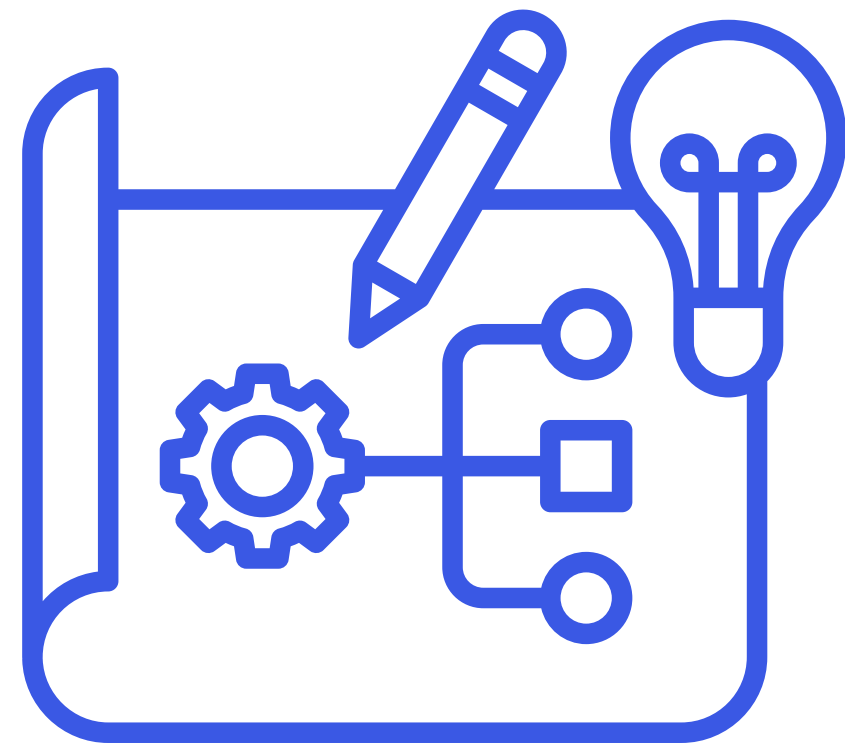
Research Question

*Can we predict whether a squirrel will **approach** or **avoid** humans based on its **behaviour**, **location**, and **time of observation**, and which features are most influential?*

OBJECTIVES

- Handle severe class imbalance.
- Identify key features via correlation analysis.
- Compare predictive models (Decision Tree vs. KNN).
- Discover natural behavioural groupings via clustering.





Research Method

2018 CENTRAL PARK SQUIRREL CENSUS

Preprocessing Pipeline

Dataset Challenges

5.9% of squirrels approached - severe imbalance

4 columns had >50% missing values. Risked unreliable patterns.

Mostly binary features with a nominal target variable

two datasets need merging to combine behavioral and environmental information

Preprocessing Decisions

Remove

Dropped high-missingness columns (>50%) and low-signal identifiers

Impute

Mode imputation for 3 low-missingness columns, row deletion would worsen imbalance.

Feature Engineering

3 composite scores: Activity, Vocal, Human Interaction Score

Data Integration

Merged hectare.csv on Hectare + Shift keys for environmental context

The “6% problem”: addressing imbalance

3.2

5.9%

POSITIVE CLASS (APPROACHES)

WHY UNDERSAMPLE?

A naive model would achieve 94% accuracy simply by guessing "No squirrel will ever approach."

Method: Stratified Undersampling selected to create a balanced training set while preserving fur color proportions

Insight: This forced the model to actually learn the patterns from both approach and non-approach cases rather than relying on frequency.

Predictive Models

DECISION TREE VS KKN

Decision Tree: Interpretable model with visible decision rules.

- Information Gain criterion used for splitting
- Class weighting reduced bias toward majority class
- Maximum depth = 5 to reduce overfitting on small training data

KKN: Distance-based, complements rule-based approach; no linearity assumption

- Features scaled using min-max normalisation
- pipeline to prevent data leakage during cross-validation
- Optimal k selected using 5-fold cross-validation on F1-score

Clustering Method

PURPOSE

Identify natural behavioural groupings without using the Approaches label.
15 behavioural and interaction features used.

KMEANS (PRIMARY)

- efficient, scalable, interpretable centroids
- k selected via Elbow Method inflection
- Silhouette Score computed for cluster quality

HIERACHICAL (VALIDATION)

- ward linkage on 200-observation sample
- confirms cluster structure via dendrogram
- ARI computed to measure similarity with KMeans.

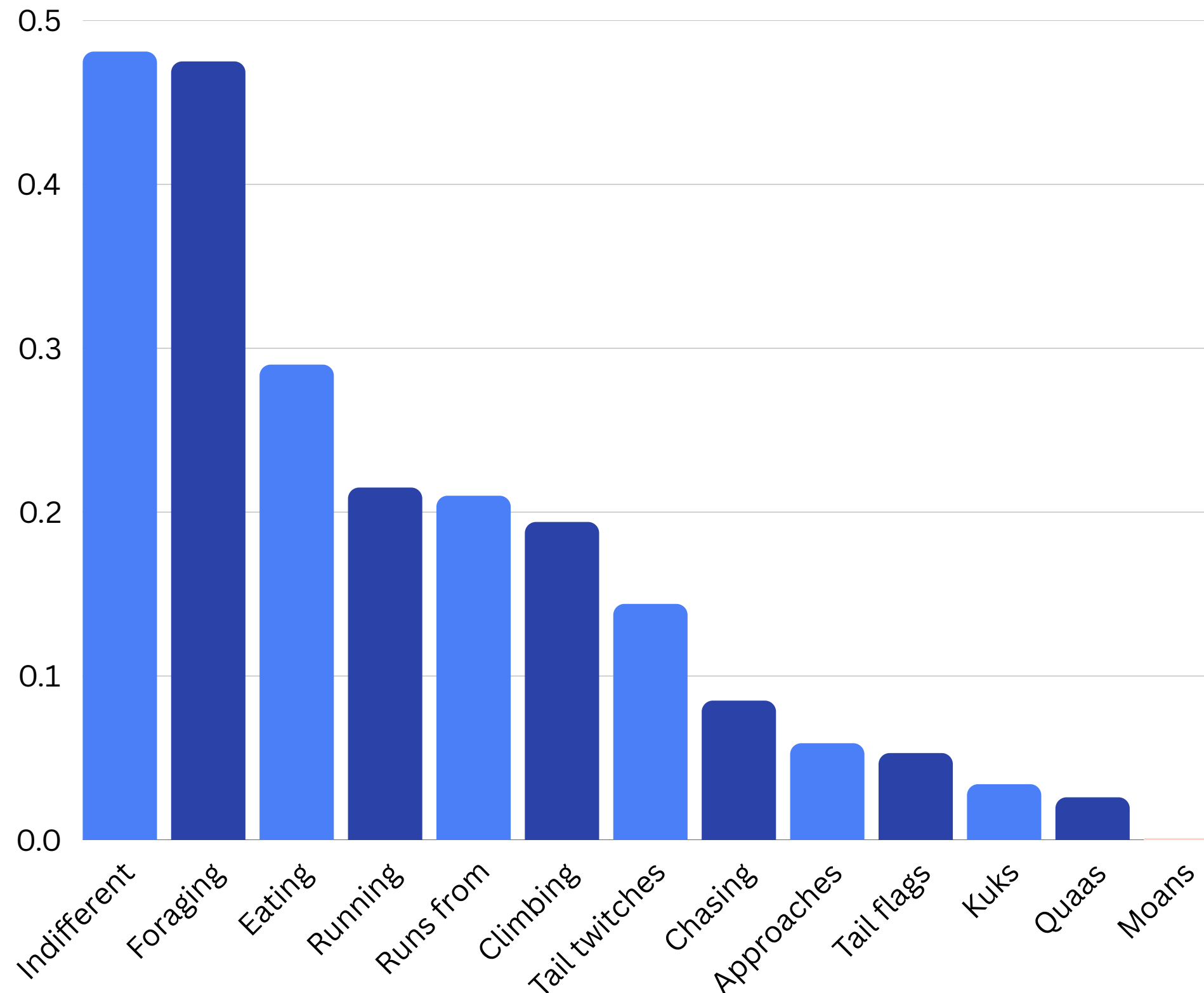


Result Analysis

2018 CENTRAL PARK SQUIRREL CENSUS

Dataset Overview & EDA

4.1



01 MOST COMMON BEHAVIOUR

Indifferent (48.1%)
Foraging (47.5%)

02 ALARM BEHAVIOURS WERE RARE

Tail twitches (14.4%)
Kuks (3.4%)
Moans (0.1%)

03 Target Variable

178 squirrels approached humans (5.9%)

04 Coordinates

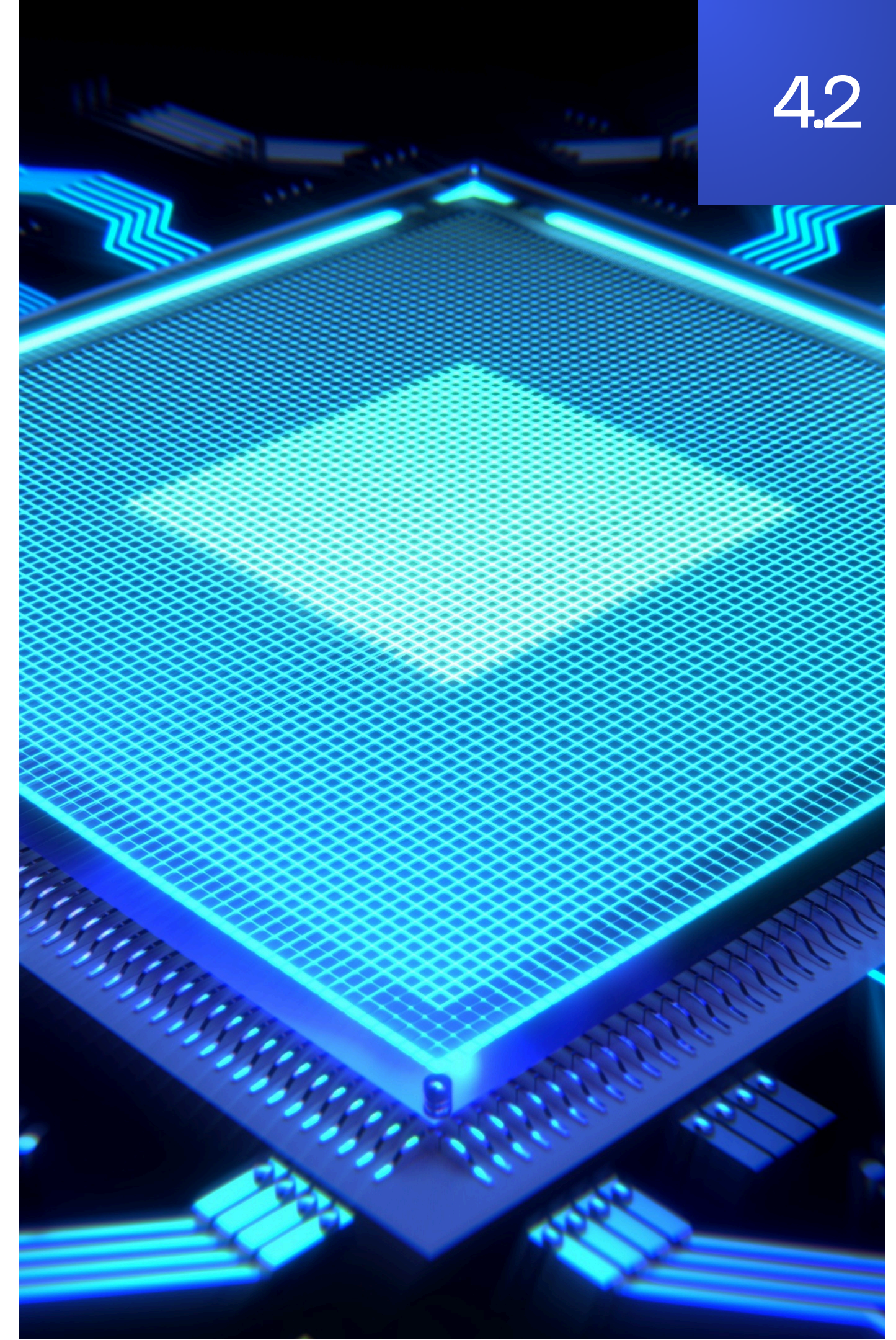
Spatial coordinates alone are weak predictors ($r = -0.052$) but hectare location is stronger ($r = 0.233$)

Preprocessing Impact

PREPROCESSING COMPARISON

Strategy	Accuracy	F1	Recall	Precision
Naive (no balancing)	0.936	0	0	0
Class weight balanced	0.795	0.326	0.833	0.203
Stratified undersampling	0.694	0.694	0.694	0.694

Naive model looks great on accuracy but predicts zero approachers — stratified undersampling gives the most balanced result.



01



STRONGEST NEGATIVE PREDICTORS

Human Interaction Score ($r = -0.507$)
Indifferent ($r = -0.467$)
less avoidance = more likely to
approach

02



STRONGEST POSITIVE PREDICTOR

Hectare location ($r = 0.233$) –
which part of the park matters,
but exact coordinates don't

03

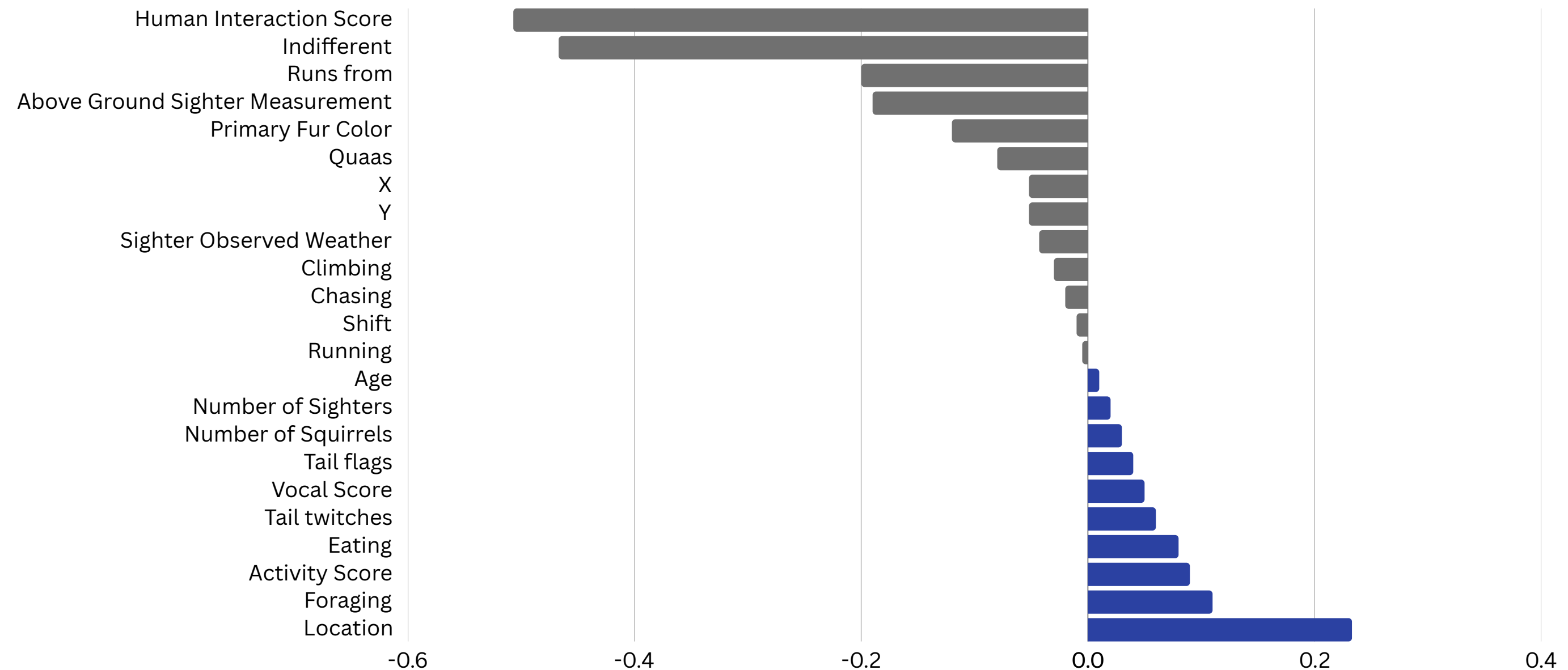


ALARM BEHAVIOURS CLUSTER

Tail twitches and vocalisations
strongly correlated ($r = 0.79$) –
squirrels showing one alarm
behaviour tend to show all

4.3

Correlation Analysis



MODEL 1: DECISION TREE (DEPTH=5)

Interpretable, uses Information Gain.

Root split:

Human Interaction Score ≤ 0.5 →
biologically meaningful

MODEL 2: KNN (K=20)

Distance-based, min-max normalised.

Optimal k selected via 5-fold cross validation

KEY TAKEAWAY

KNN outperforms on all metrics — especially recall, which is the most important metric for catching true approachers

Supervised Learning

DECISION TREE VS KNN (K=20) PERFORMANCE



Model	Accuracy	F1	Recall	Precision
Decision Tree (depth = 5)	0.694	0.694	0.694	0.694
KNN (k = 20)	0.764	0.764	0.75	0.771

Model Evaluation

4.4

KNN WINS ON PERFORMANCE

F1 = 0.764, Recall = 0.750

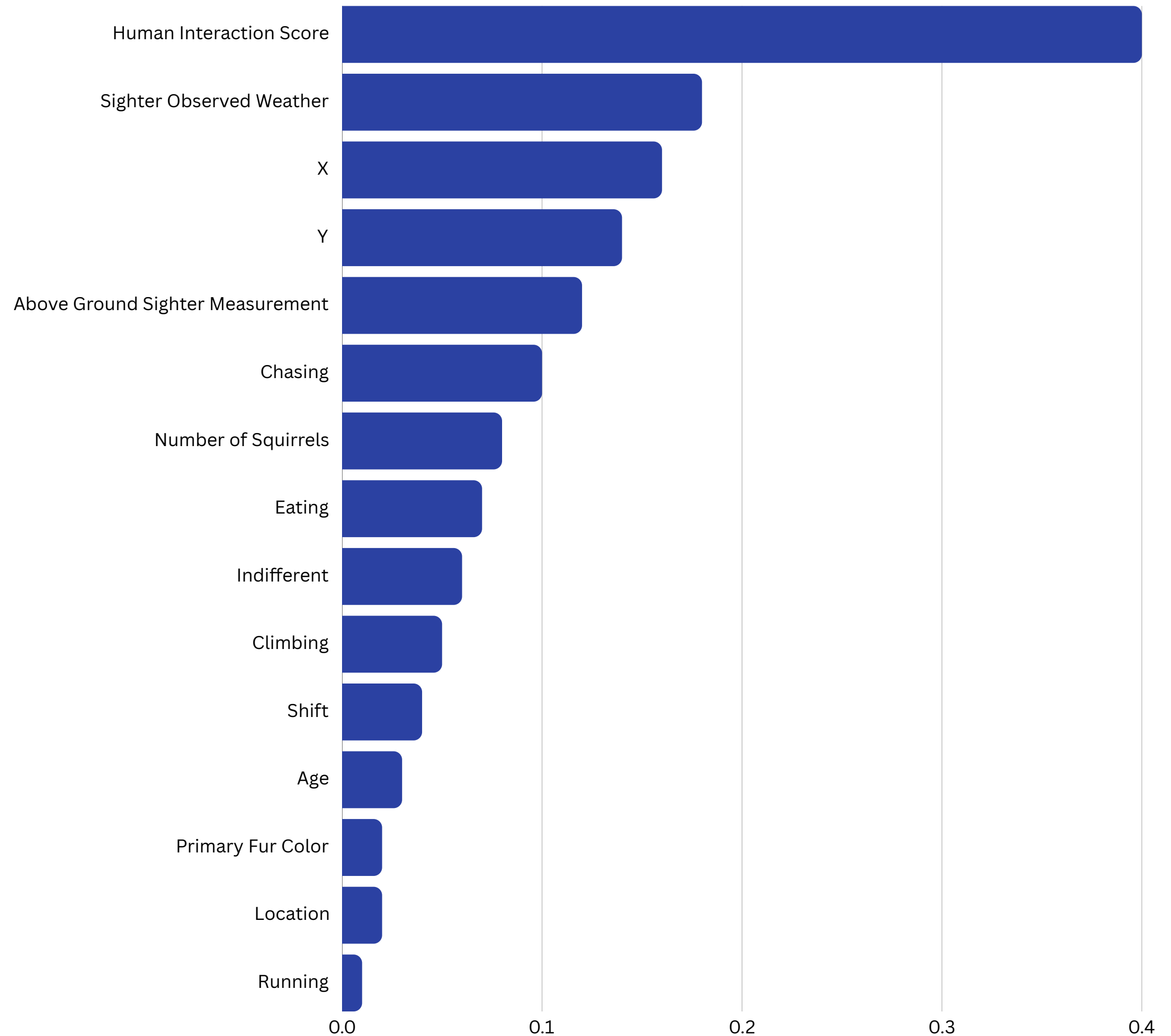
DT WINS ON INTERPRETABILITY

Root split is biologically meaningful

BOTH MODELS CONSISTENT

CV results:

- DT 0.733 ± 0.031
- KNN 0.765 ± 0.040



Model Evaluation

4.4

KNN WINS ON PERFORMANCE

F1 = 0.764, Recall = 0.750

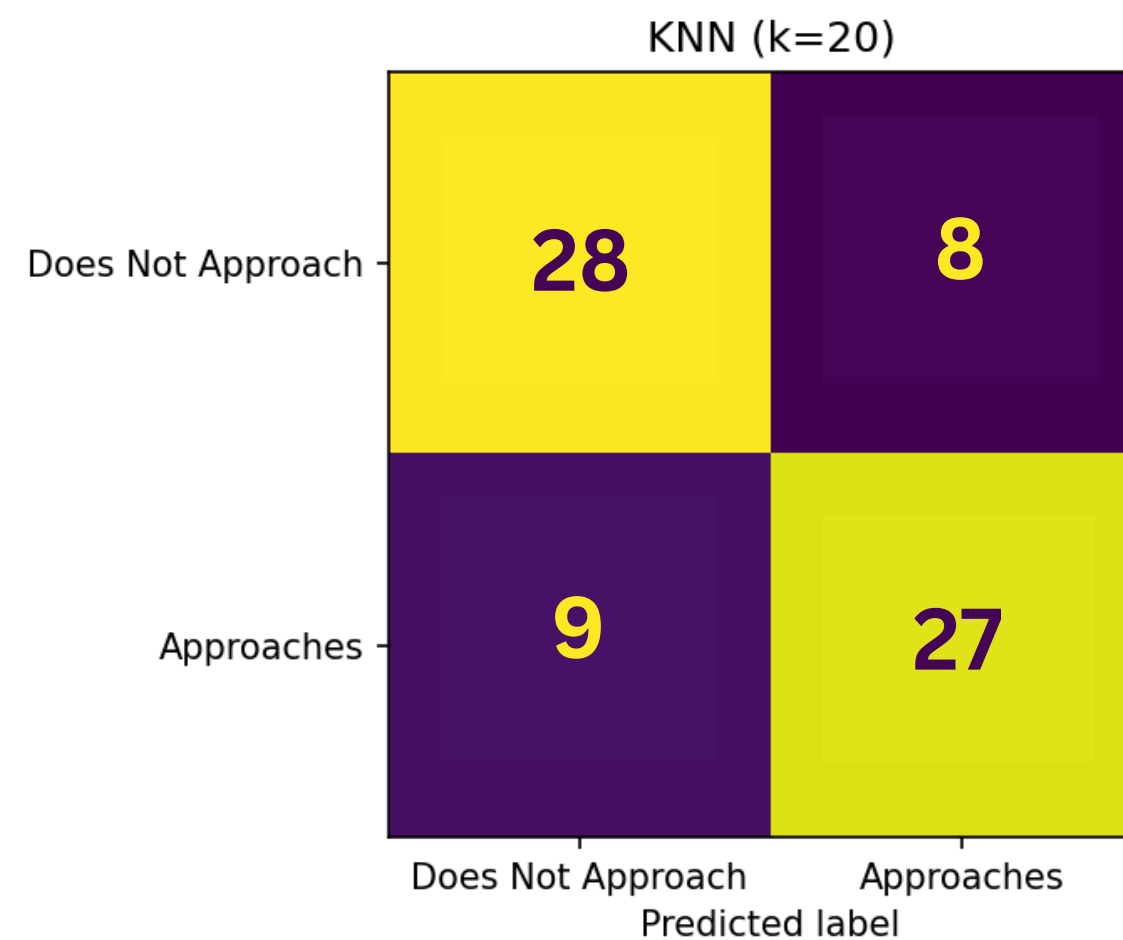
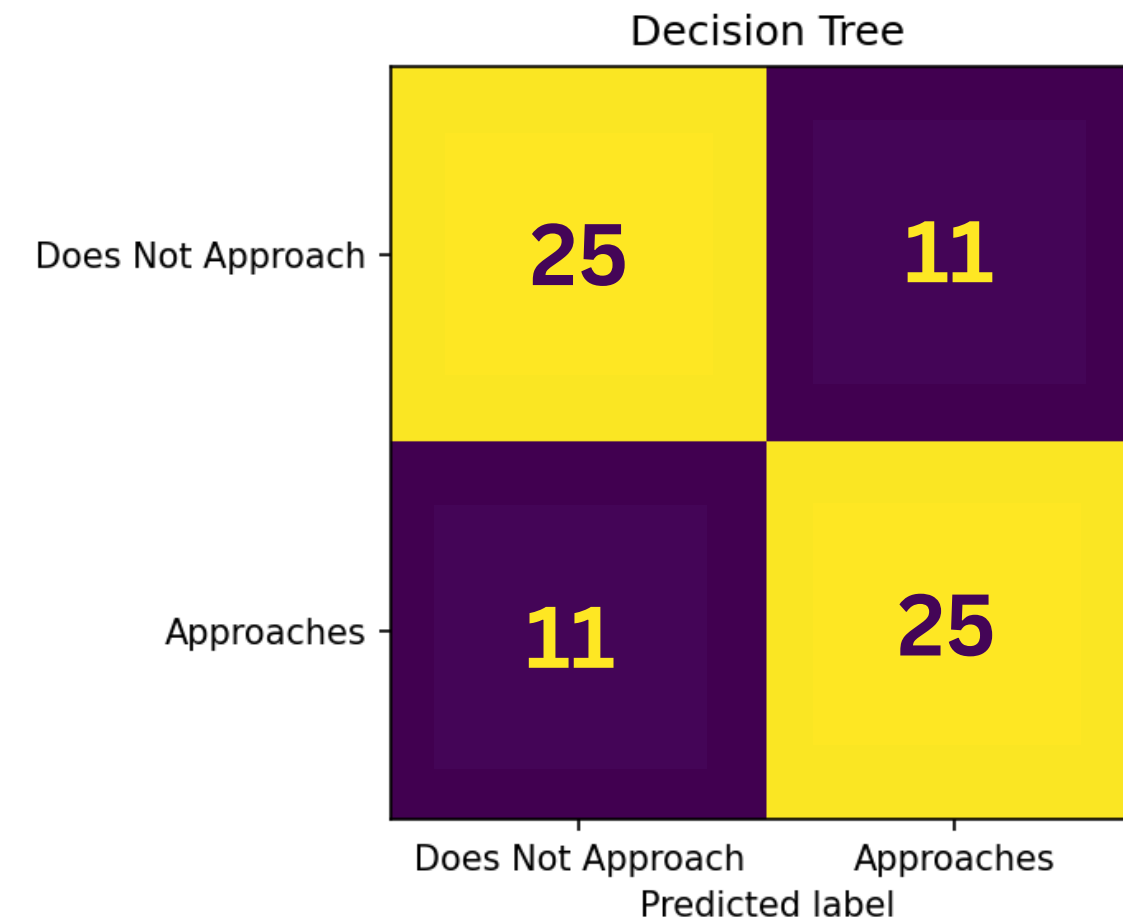
DT WINS ON INTERPRETABILITY

Root split is biologically meaningful

BOTH MODELS CONSISTENT

CV results:

- DT 0.733 ± 0.031
- KNN 0.765 ± 0.040



15.5% **Human-Approachers**
Low alarm, low avoidance

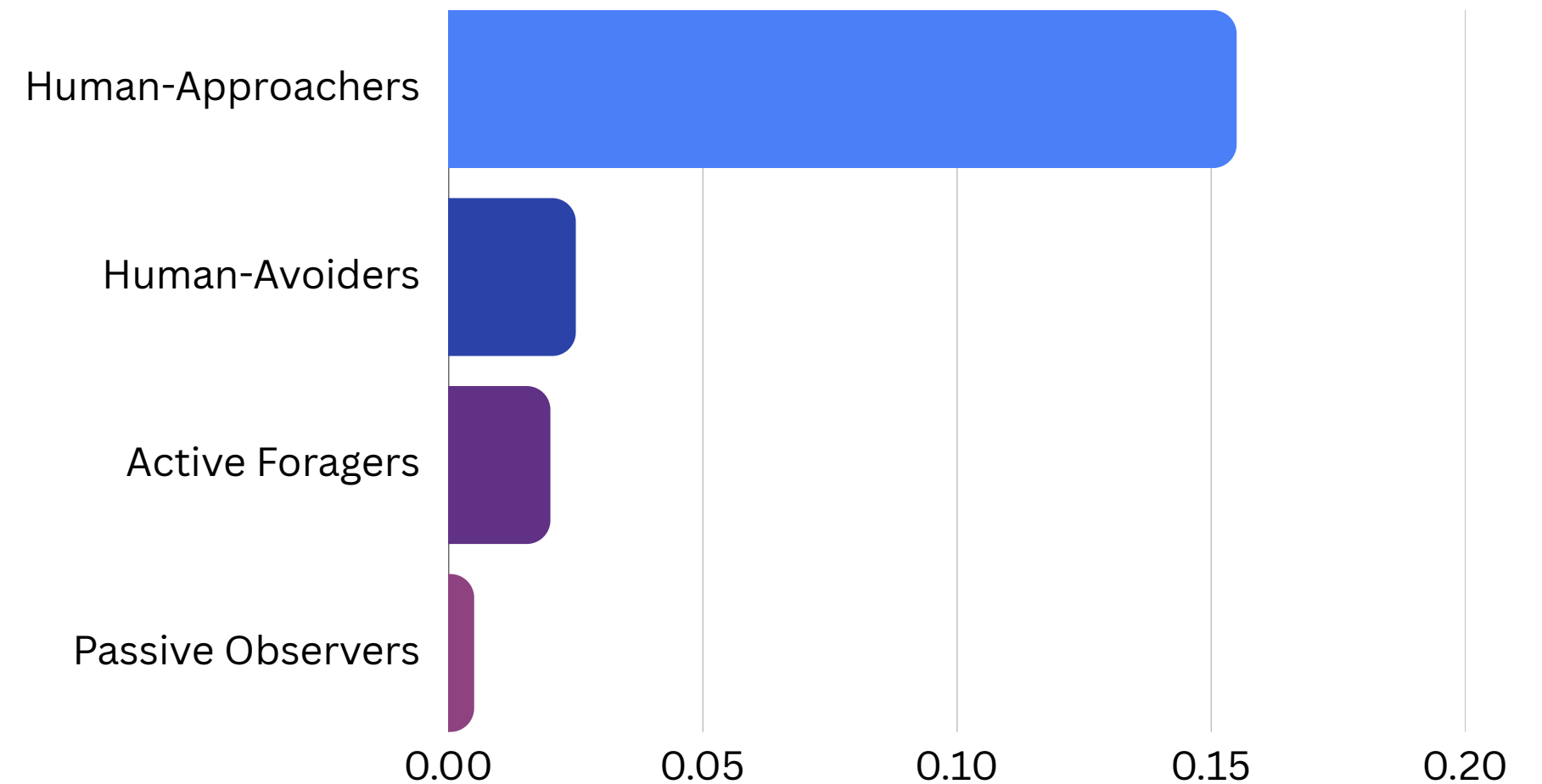
2.5% **Human-Avoiders**
High vocal, runs from = 100%

0.5% **Passive Observers**
High indifferent, stationary

2.0% **Active Foragers**
Food-focused, unbothered

Clustering Result

THE 4 CLUSTERS ARE THE MAIN STORY ↗



Baseline approach rate is only 5.9%

Human-Approachers at 15.5% is more than double, confirming a distinct human-tolerant group exists



Insights Summary

2018 CENTRAL PARK SQUIRREL CENSUS

5.9%
→ 15.5%

Distinct Group Confirmed

Human-Approachers cluster approach rate vs dataset baseline more than double, confirming a distinct human-tolerant group exists

Human Interaction Score Ranked Strongest Predictor

$r = -0.507$
DT = 0.4

Human Interaction Score dominated Pearson ($r = -0.507$), NMI (≈ 0.25), and DT importance (≈ 0.40), disposition toward humans drives approach, not activity or location

KNN's F1:
0.764

Performance vs Interpretability

Outperformed Decision Tree (F1 = 0.694) across all metrics, but DT remains valuable for interpretability

Strategies for Class Imbalance

94% accuracy →
0 positive cases

"Naive identified zero approachers and class weighting overcorrected with 80% false positives only stratified undersampling balanced precision and recall, producing a genuinely useful model."



Findings Report

2018 CENTRAL PARK SQUIRREL CENSUS

Preprocessing Decisions

PREPROCESSING DECISIONS AND THEIR IMPACT

94%

Naive Model Accuracy

0% recall on positive class due to majority class dominance (94.1%)

0.833

Class Weighting

Class Weighting helped recall but created too many false positives 80% of positive predictions were wrong

≈0.69

Stratified Undersampling

Most balanced result, equal precision and recall forced model to learn from a balanced feature space

Feature Engineering

- Combining Indifferent + Runs from into a single Human Interaction Score made it the dominant Decision Tree predictor (≈0.40 importance), capturing more signal than either variable alone.

NMI Anomaly

Weather conditions showed high NMI (0.671) but near-zero Pearson correlation ($r = 0.043$), confirming no linear relationship. This gap reveals that NMI values from discretised continuous variables cannot be interpreted without cross-referencing other measures.

Correlation Findings

01 Human Interaction Score

$r = -0.507$ (Pearson)
NMI ≈ 0.25
DT importance ≈ 0.40

02 Alarm Behavior Cluster

$r = 0.51$
TAIL FLAGS vs VOCAL SCORE

$r = 0.79$
TAIL TWITCHES vs VOCALISATION

03 Approaches vs Indifferent

$r = -0.47$
NOT OPPOSITES

04

CONFOUNDERS

- Food provisioning
- Observer behaviour
- Individual boldness

01

HUMAN INTERACTION SCORE

#1 across all three methods, confirming that disposition toward humans drives approach behaviour more than activity or location.

02

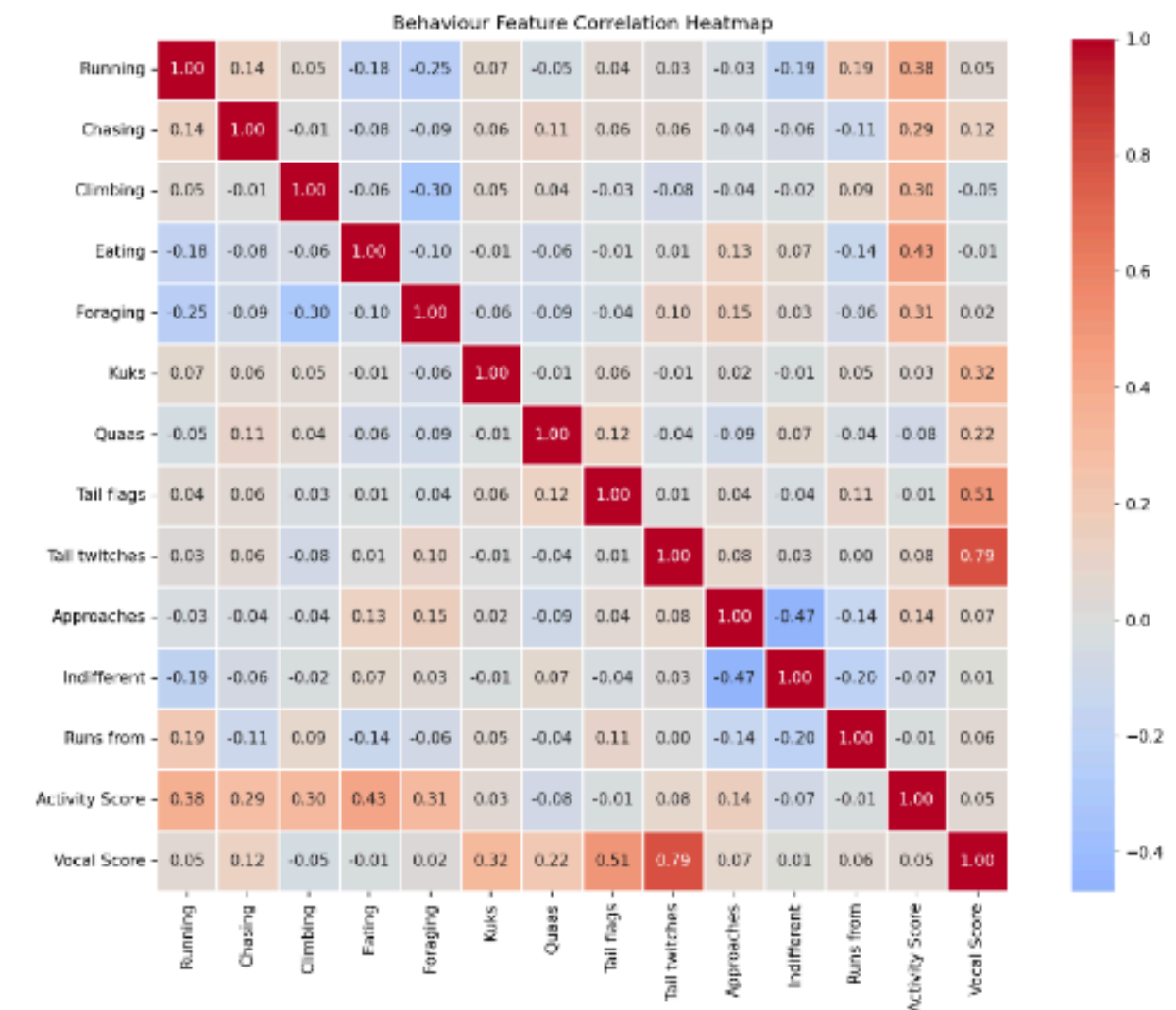
ALARM BEHAVIOUR CLUSTER

Alarm behaviours form a coordinated response system, tail twitches correlate strongly with vocalisation ($r = 0.79$) and tail flags with vocal score ($r = 0.51$), meaning squirrels displaying one alarm behaviour likely display others simultaneously.

03

APPROACHES vs INDIFFERENT

Not opposites, a squirrel can be neither approaching nor indifferent; it may instead be foraging or climbing. This multidimensionality justifies clustering, which captures behaviour combinations that pairwise correlation alone cannot reveal.



Model Evaluation and Comparison

06

KNN

KNN leads on all metrics, higher accuracy (0.764), recall (0.750), and precision (0.771), making it the stronger classifier for identifying true approachers.

Decision Tree (DT)

Decision Tree achieved equal precision and recall (both 0.694), consistent and balanced, but less powerful overall.

Model	Accuracy	F1	Recall	Precision	CV F1 $\pm$ std
Decision Tree (depth=5)	0.694	0.694	0.694	0.694	0.733 $\pm$ 0.031
KNN (k=20)	0.764	0.764	0.75	0.771	0.765 $\pm$ 0.040

Conclusion KNN v.s. DT

Since recall matters most for our research question, missing a true approacher is costlier than a false alarm, KNN holds the practical advantage despite both models being evaluated on the same balanced test set.

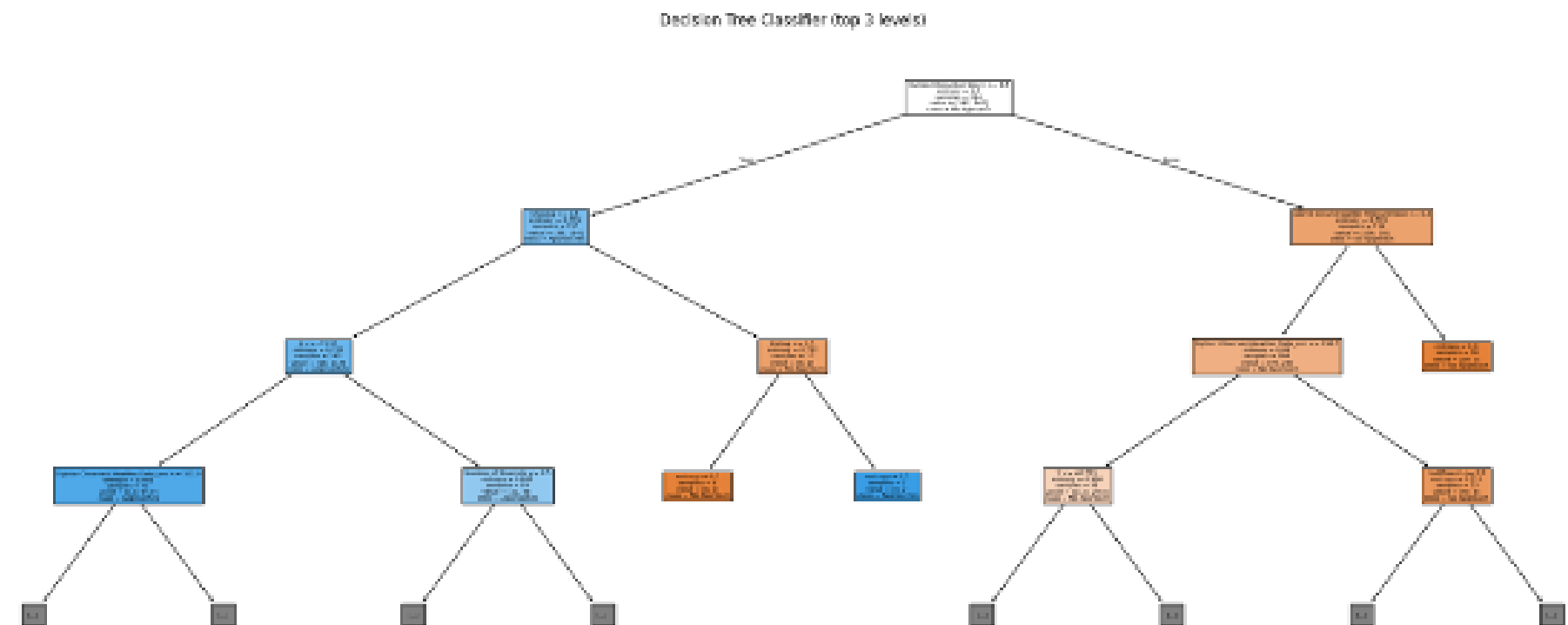
Model Evaluation and Comparison

KNN

Classifies purely through neighbourhood similarity across 25 dimensions, matches instances without revealing which features matter most or why.

Decision Tree

Root split at Human Interaction Score ≤ 0.5 is biologically meaningful; squirrels with zero human interaction events are routed to the approach-positive branch, exposing which features drive predictions.

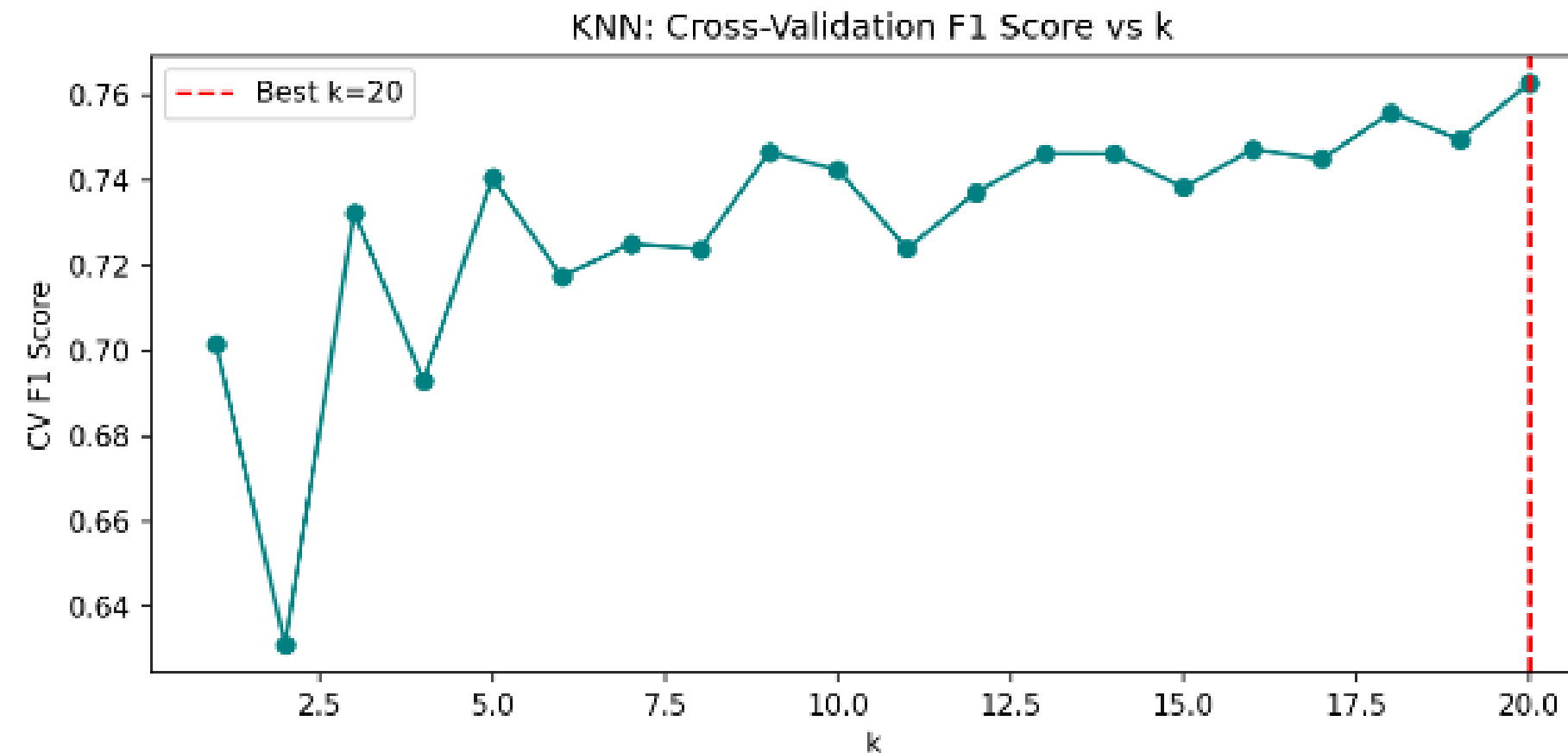


Conclusion On Interpretability

For a research question that depends on understanding which behaviours drive approach, the Decision Tree's interpretability is a valuable complement to KNN's superior metrics.

Model Evaluation and Comparison

06



KNN's Curse of Dimensionality

KNN relies on distance measurements between points, which become unreliable as dimensions increase, with 25 features and only 284 training points, the unusually high $k=20$ reflects this compensation, averaging over more neighbours to stabilise noisy distance estimates. The Decision Tree is unaffected as it splits purely on information gain at each node, making dimensionality irrelevant to its performance. Applying PCA prior to KNN to reduce dimensionality could make distance estimates more reliable and improve recall further.

Model Evaluation and Comparison: Alternative Models

Random Forest

Builds many trees and averages their predictions, reducing variance from our small positive-class sample while keeping feature importance scores interpretable. The best next step is to **maximise recall**.

Logistic Regression

Offers interpretable coefficients that quantify exactly how rising Human Interaction Scores shift approach probability. However it assumes feature independence and linearity. Best if **interpretable effect sizes are the priority**.

Gradient Boosting

Likely achieves the highest raw performance through sequential error correction. However overfitting risk is substantial given only 284 training points without careful regularisation tuning.

Human-Approachers

calm, no fear signalling, consistent with habituation. Distinct group confirmed without using Approaches label as input

Human-Avoiders

high activity reflects escape, not foraging threat-response behaviour

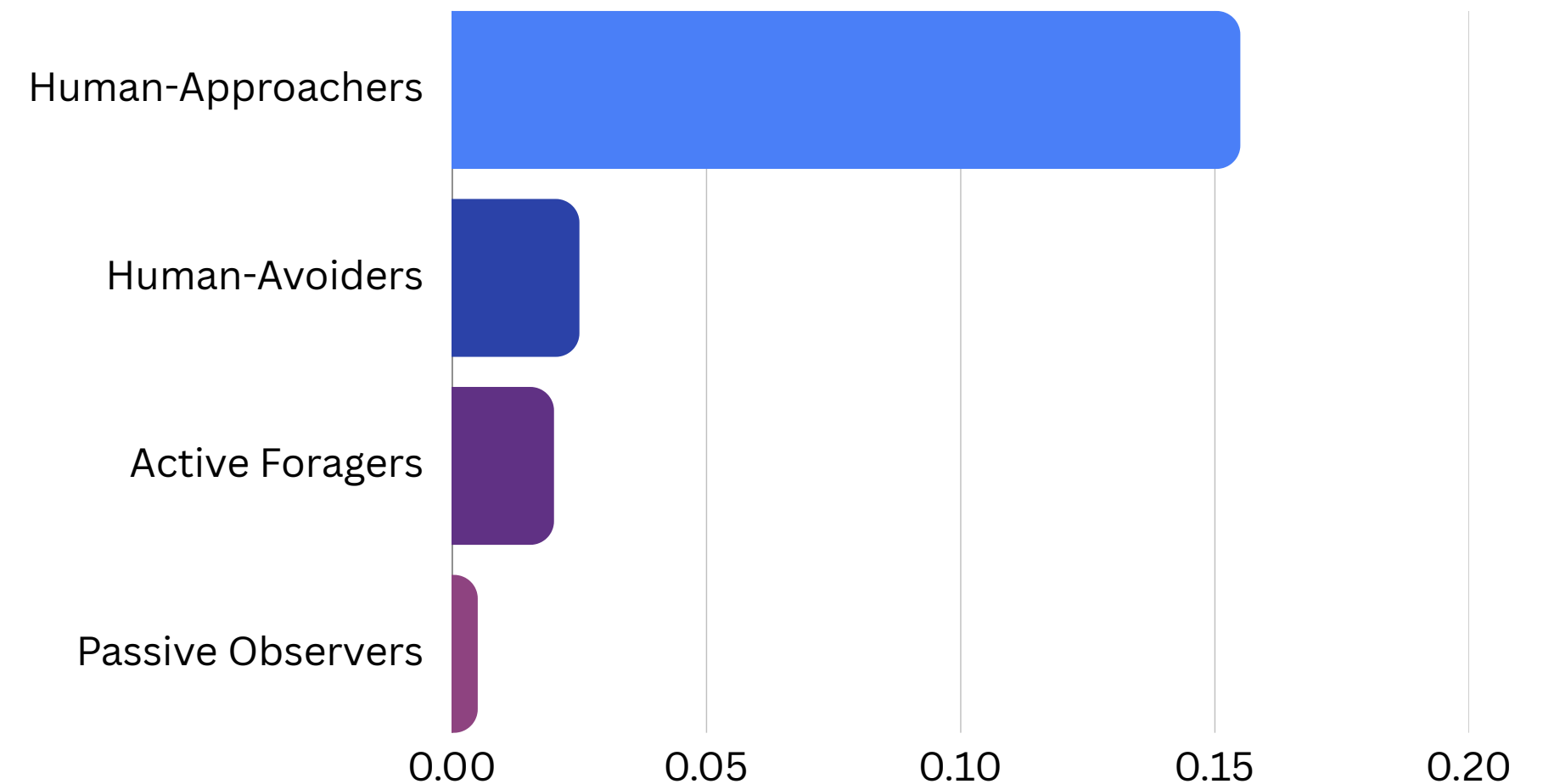
Passive Observers

tolerate humans without engaging or avoiding, watchful and stationary

Active Foragers

food-focused and indifferent, human presence simply doesn't register

Clustering Interpretation



k=4 was confirmed by both the Elbow Method and dendrogram, though the monotonically increasing Silhouette Score could not independently confirm this. The PCA projection explains only 37.9% of variance, meaning true cluster separation is likely stronger in higher dimensions than the 2D visualisation suggests. An ARI of 0.345 indicates moderate agreement between KMeans and Hierarchical Clustering, providing reasonable but not perfect validation of the four-cluster structure."

Research Limitations

- » **Data Constraints:** Cross-sectional (Autumn 2018 only).
 - Repeated sightings possible (no individual tags).
- » **Observation Bias:** Volunteer variability (movement speed, distance).
- » **Technical Limitations:** High variance due to small positive class (178 records); potential artifacts in continuous variable binning.



Conclusion & Insights

BEHAVIORAL PLASTICITY MATTERS MOST

A squirrel's learned behavior, whether it actively engages with humans or avoids them, has a stronger influence than its physical location in the park.

01

02

PREDICTION VS. UNDERSTANDING

KNN delivered stronger predictive accuracy, while Decision Trees provided clearer biological insight into why squirrels approach humans.

EVIDENCE OF HABITUATION

The clear "human-approachers" cluster suggests that urban squirrels may be developing learned tolerance toward humans over time.

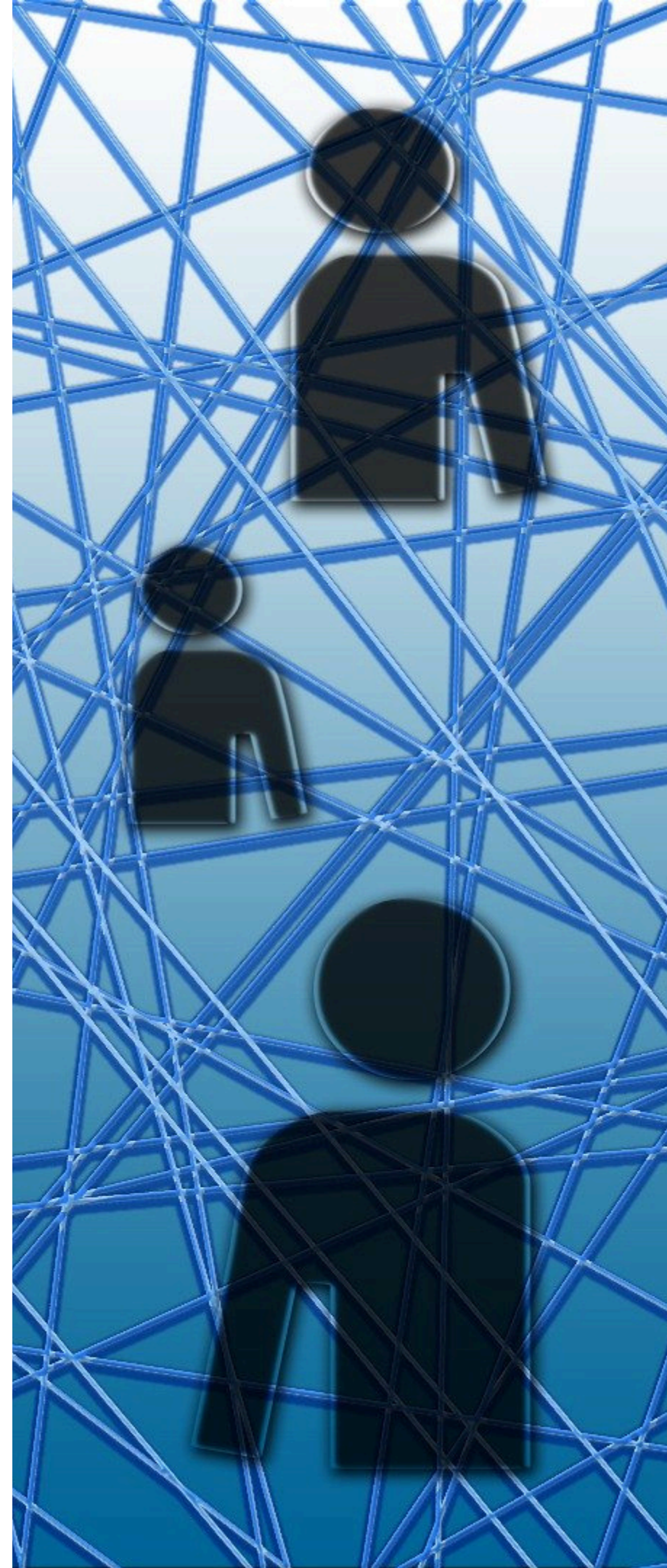
03

04

IMPROVING FUTURE RESEARCH

Future wildlife studies should move beyond simple sighting counts and focus on long-term tracking of individual animals to better understand habituation.

Thank You!



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